

Introduction to the World of Artificial Intelligence

The Generative AI Revolution

Session 4 | From Pattern Recognition to Creating Text, Image, Audio & Code

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Session 4 Roadmap

Recap of Previous Session & Today's Objectives

Session 3 Recap

Brain Inspiration: Function of neurons and network layers

Deep Learning: Discovering complex patterns by adjusting weights

Session 4 Objectives

Concept of Generative AI: Distinguishing discriminative AI from creative AI

The Secret of Language Models: How chatbots work in simple terms

Understanding Tools: Practical comparison of leading text and image models

Hands-on Experience: Live content generation and prompt engineering principles

Fundamental Definition

What is Generative AI? (Generative Artificial Intelligence)



Generative AI (Creative)

Core Task: Creating entirely new and original content

Visual Example: Painting an astronaut cat from scratch

Text Example: Writing a full formal letter or creative story



Traditional AI (Discriminative)

Core Task: Classifying and detecting existing data

Visual Example: Detecting whether a cat is present in a photo

Text Example: Identifying spam marketing emails

The Engine Behind Chatbots

How Do Large Language Models (LLMs) Actually Work?



Reading Billions of Pages

Trained on books, research papers, and websites worldwide

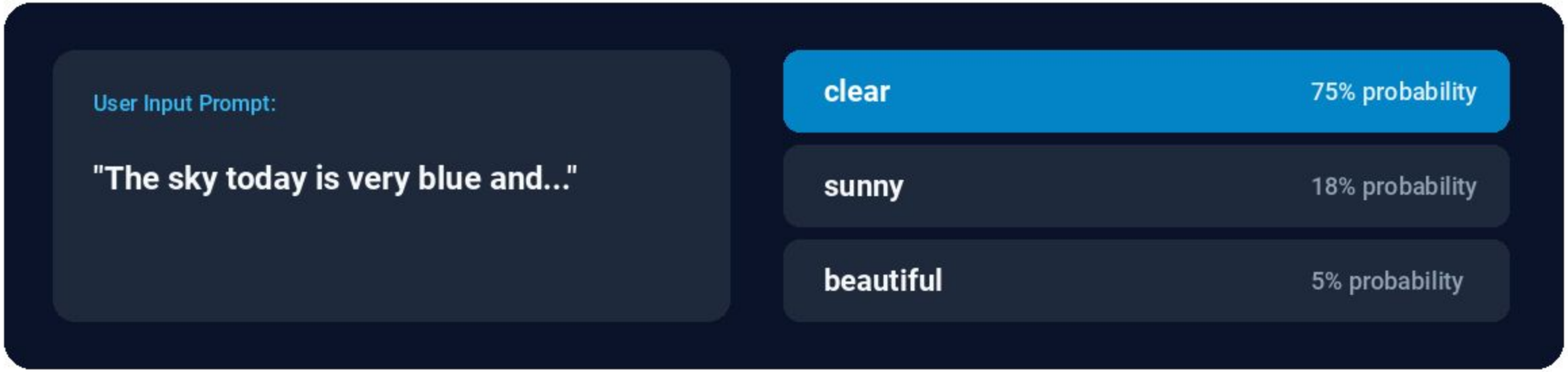
Discovering Language Structure: Learning semantic connections between words and sentences

Not Rote Memorization: Understanding writing patterns rather than copying text

Multilingual Mastery: Capable of conversing and reasoning across dozens of languages

The Core Secret in Simple Terms

The core idea behind LLMs: Predicting the next word!



1. Advanced Auto-Complete

Intelligently predicts the next word, like modern smartphone keyboards.

2. Probability Calculation

Selects the optimal word based on the full conversational context.

3. Continuous Chaining

Repeats this loop word by word to synthesize full paragraphs and essays.

Intelligent Writing Assistant

Everyday Applications of AI Text Generation



Brainstorming & Ideation

Suggesting brand names, ad scenarios, and travel itineraries



Summarization & Translation

Condensing long reports into clear, key takeaways



Writing & Content Creation

Drafting business emails, articles, and social media posts

Turning Words into Images

Image Generation with Generative AI



Prompt: "A glowing crystal castle floating on clouds in sunset light"

Design from Description: Directly converting sentences into photos and paintings

Diverse Artistic Styles: Rendering as realistic photos, watercolor, or digital art

Smart Editing: Changing backgrounds or adding objects to existing photos

Beyond Text and Photos

Generating Video, Audio, and Code



Software Code Generation

Writing and debugging code rapidly for websites and apps



Voice & Music Generation

Natural voiceover from text and composing custom music tracks



Video & Animation

Creating cinematic video clips from text or still photos

Knowing the Major Players

Comparing Leading Generative AI Models

Model / Tool	Developer	Key Strength & Primary Use
ChatGPT (GPT)	OpenAI	All-rounder in conversation, ideation, and ecosystem plugins
Gemini	Google	Multimodal understanding (text, video, image) & Google Workspace integration
Claude	Anthropic	Highly nuanced natural writing, precise coding, and long document analysis
Midjourney / Imagen	Image Specialists	Stunning artistic visuals and high-detail photorealistic imagery

Smart Tool Selection

Practical Comparison: Which Model Fits Which Task Best?

Practical Limitations

Free Tier Limits: Daily message caps on basic tiers

Hallucinations: Potential for incorrect answers delivered with confidence

Image Detail Flaws: Occasional artifacts and errors in rendering in-image text

Practical Strengths

Persian Writing & Coding: Clear edge in Claude and GPT

Live Search & Large Files: High speed with Gemini

Marketing & Art Imagery: Stunning quality with Midjourney

Practical Workshop 1

Live Demo: Intelligent Text & Content Generation

Ready-to-Test Classroom Prompt:

"You are a business consultant. Write a 3-day launch plan for a local book cafe's Instagram page in a warm, friendly tone."

Live Demonstration Steps by Instructor:

Step 1: Send a short, vague prompt and evaluate the generic response

Step 2: Send the detailed prompt above with role, context, and tone

Step 3: Request formatting the output into a structured schedule table

Practical Workshop 2

Live Demo: Creating Images from Text Descriptions



The Golden Formula for Image Prompts:

Subject + Environment & Background + Art Style + Lighting

Experiment 1: Generating a modern library with turquoise neon lighting

Experiment 2: Changing the style of the same image to watercolor or animation

Key Takeaway: Precise details dramatically improve output quality

Professional Best Practices

Three Golden Rules of Generative AI



1. Precise Prompting

Clearly define role, goal, audience, and output format



2. Fact-Checking Outputs

Verify statistics, figures, and technical data via human review



3. Protecting Privacy

Never share passwords, sensitive personal data, or confidential company files

Discussion & Exchange

Interactive Activity: Your Experience with Generative AI



"What is the most interesting or surprising response you have ever received from chatbots or image tools?"

Answer this question: Which time-consuming daily task would you most like to delegate to AI?

Final Review

Session 4 Summary

1. Create Instead of Detect: Generative AI can produce new text, images, audio, and code.

2. Language Model Secret: Chatbots build coherent sentences by intelligently predicting the next word.

3. Specialized Tool Diversity: Each model (GPT, Gemini, Claude) has unique strengths.

4. The Power of Good Prompts: Output quality directly depends on prompt precision.

5. Human Oversight: Verifying factual accuracy and guarding confidential data remains essential.

Addressing Concerns

Frequently Asked Questions About Generative AI

Which tool is completely free to start with?

Basic versions of ChatGPT, Gemini, and Claude are free and readily accessible.

Is generated content copied from the internet?

No, words and pixels are synthesized in real time from scratch.

Will AI replace our jobs?

People who master AI will replace those who do not use it.

Weekly Practical Challenge

Session 4 Assignment: Your First Creative Project!



Create & Share

Hands-on experience with one of the featured tools

1. Practical Text Generation: Use a chatbot to write a plan, email, or content relevant to your work.

2. Creative Image Generation: Craft a compelling image with a detailed prompt in an image generation tool.

3. Share in Course Group: Post your prompt and final result to discuss with fellow participants.

End of Session 4 | Introduction to Artificial Intelligence

Thank you for your participation and engagement!

Next Session (Session 5):
Everyday AI Tools

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Image Sources

https://media.easy-peasy.ai/1de4a307-14e8-4a21-ac10-6a554f139599/43e55c27-8c75-4e0b-bdc7-2168df919498_medium.webp

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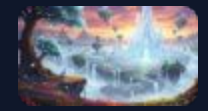
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